الجامعة السورية الخاصة
SYRIAN PRIVATE UNIVERSITY

Faculty of Artificial Intelligence Engineering
Department of Robotics and Intelligent Systems

# Remote Control of the Motion of a Differential Drive Robot by Moving the Palm of the Hand

**Semester Project Report**
**Submitted to the Department of Control Engineering and Robotics in Partial Fulfillment of the Requirements for the Semester Project Course**

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Hand Gesture Control
Left
Right
Backward
HC-05 Bluetooth
Arduino Nano
Caster Wheel
Gear Motors

## 1. Project Definition

This project aims to develop a system for controlling the movement of a mobile robot using hand gestures. Hand movements are translated into control commands that are wirelessly sent to the robot via Bluetooth. The robot receives these commands and adjusts its motion accordingly, allowing it to move forward, backward, or turn in various directions.
This approach provides a simple and intuitive method for controlling the robot through natural hand gestures, offering a smoother and easier alternative compared to conventional systems.

- **Project Objective**: To develop an intuitive control interface that is user-friendly, especially for elderly and physically impaired users .
- **Innovation**: The integration of hand gesture recognition to interact with the robot offers a more natural and personalized human-robot interaction experience .

## 2. Project Objectives

The objectives of this project are:

1. **Developing a Gesture-Based Control System:** The project aims to develop a control interface using hand gestures to enable the user to direct the robot in a natural and intuitive manner.
2. **Implementing Wireless Communication Technologies:** Using Bluetooth technology, commands will be transmitted wirelessly between the transmitter and receiver unit, allowing remote control of the robot.

3. **Designing a Simple and Efficient Mobile Robot:** The project focuses on designing a robot capable of executing commands accurately and reliably using DC motors and a differential drive system.
4. **Integrating Multiple Electronic Technologies into a Single System:** The system involves the use of electronic components such as motion sensors, Bluetooth, and an Arduino microcontroller for seamless communication and control.
5. **Presenting a Practical Model for Robotic Applications:** The project will provide a practical model for controlling a robot using hand gestures, contributing to the development of human-machine interaction in various contexts.

## 3. Problem Statement

The primary problem addressed by this project is the challenges faced by elderly and physically impaired individuals in using conventional control interfaces. Current systems are complex and often inconvenient for people with special needs.

- **Problem 1:** Elderly and physically impaired individuals face difficulties using traditional control interfaces that rely on buttons or screens.
- **Problem 2:** There is a lack of intuitive systems for assistive human-robot interaction (HRI), limiting access to these systems.
- **Problem 3:** Limited accessibility of robotic systems for daily assistance tasks, making them underutilized in daily life.
- **Problem 4:** The need for simple, natural, and low-effort control methods to enhance quality of life for people with disabilities.

## 4. Methodology

The project is divided into several stages to ensure the development of an efficient, integrated system. The methodology followed in this project includes the following stages:

1. **System Requirements Definition and Component Selection:** In this phase, the primary system requirements were identified, and suitable electronic components, such as motion sensors (MPU-6050), Arduino Nano microcontroller, and HC-05 Bluetooth modules, were selected.
2. **Design and Development of the Gesture Sensing Unit:** The gesture sensing unit was developed using motion sensors (MPU-6050) to detect hand movement and convert the data into actionable commands. This data is later used to control the robot's movement.
3. **Implementation of Data Processing and Gesture Mapping Algorithm:** An algorithm was developed to process the data collected from the motion sensors and map the gestures to control commands (e.g., move forward, backward, left, right, stop). These commands are sent to the robot’s control unit via Bluetooth.
4. **Development of Wireless Communication Module (Bluetooth):** The wireless communication module was developed using HC-05 Bluetooth modules to transmit the commands between the transmitter and receiver units. This module ensures quick response and efficient remote control.
5. **Integration of Robot Hardware and Motor Control System:** The system was integrated with the motor control system using the L298 motor driver to control the motors that drive the robot. DC motors and a differential drive system were used to move the robot in the desired directions.
6. **System Testing, Calibration, and Performance Evaluation:** Tests were conducted to evaluate the robot's response, connection stability,

and motion efficiency. Adjustments were made based on the results to achieve optimal performance.

## 5. Block Diagram

The block diagram illustrates the overall system architecture, consisting of two main units: the **transmitter unit** and the **receiver unit**.

- In the transmitter unit, the MPU-6050 sensor measures hand movements, then the Arduino Nano processes this data and converts it into control commands transmitted wirelessly via the HC-05 Bluetooth module.
- In the receiver unit, the commands are received by the HC-05 module, and the Arduino Nano converts them into signals sent to the L298 motor driver to control the robot's movement.
  This diagram provides a clear overview of the information flow between the main system units, helping to understand the project operation.

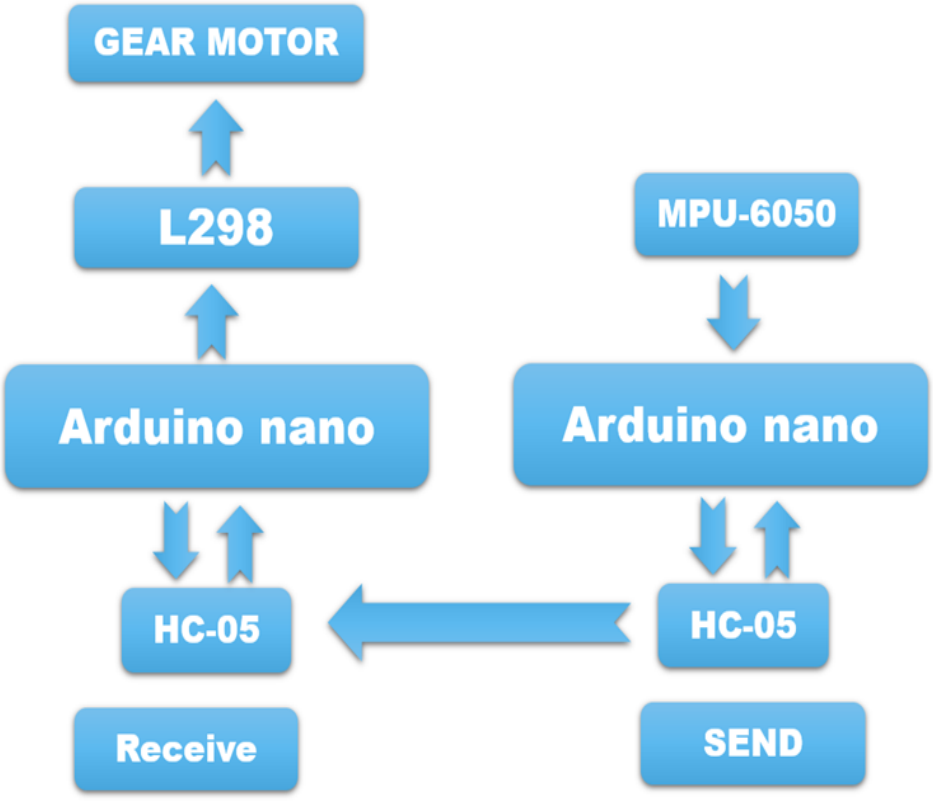


**Figure 1: Block Diagram of the Proposed System**

## 6. Flowchart

The flowchart illustrates the sequence of operations in the system from hand movement detection to robot motion.

- Data is captured from the MPU-6050 sensor and converted into digital values by the Arduino Nano.
- The values are analyzed to determine the gesture type (forward, backward, left, right, stop).
- The commands are wirelessly transmitted to the receiver unit, which directs the L298 motor driver to control the motors according to the gestures sent. The flowchart helps to understand the logical flow of the program and task distribution between the two units.

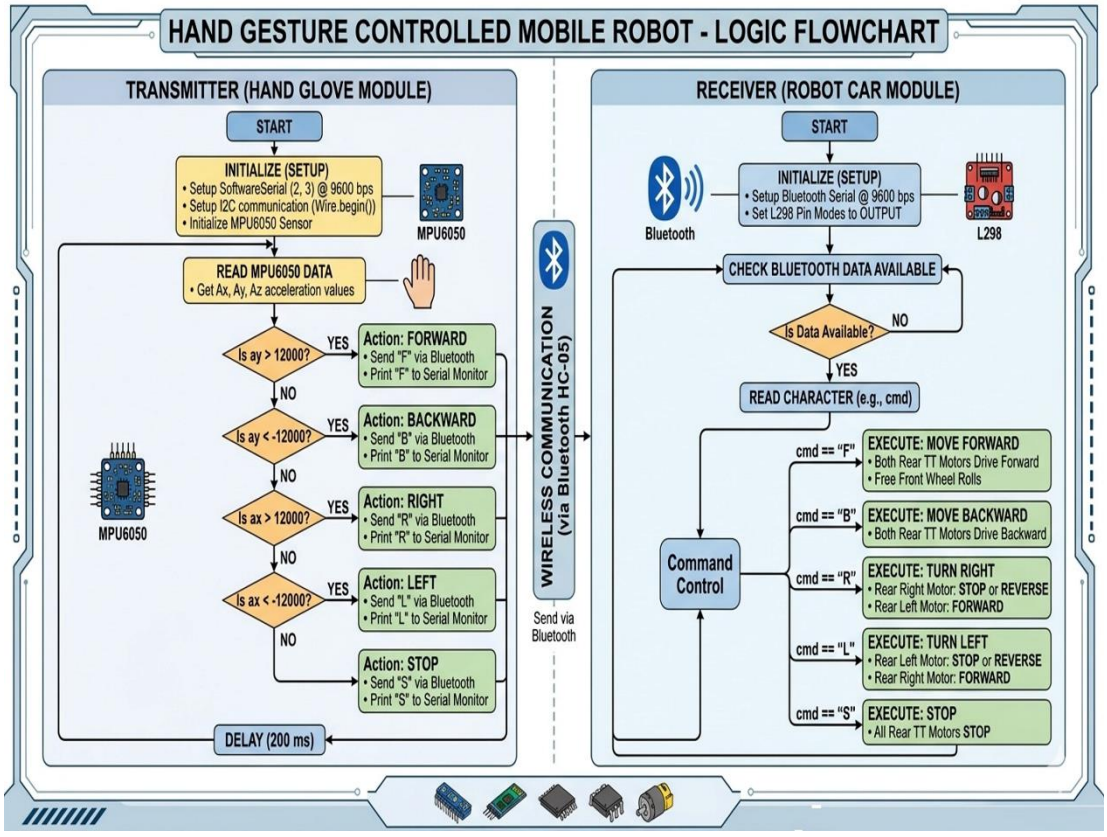


**Figure 2: Flowchart of the Gesture-Control Process**

## 7.Transmitter Diagram

The transmitter diagram illustrates the connection of electronic components responsible for capturing hand gestures and converting them into control commands that can be sent to the robot.

- The **MPU-6050 sensor** measures hand movements along the X, Y, and Z axes.
- The sensor data is processed by the **Arduino Nano**, which converts it into digital signals.
- These signals are transmitted via the **HC-05 Bluetooth module** to the receiver unit to control the robot's movement.
- The system is powered by a **rechargeable lithium battery** and an additional power source (Power Bank) to ensure stable operation. This diagram shows the internal connections between the sensor, microcontroller, and Bluetooth module, highlighting how the system is configured to send precise and rapid control commands.

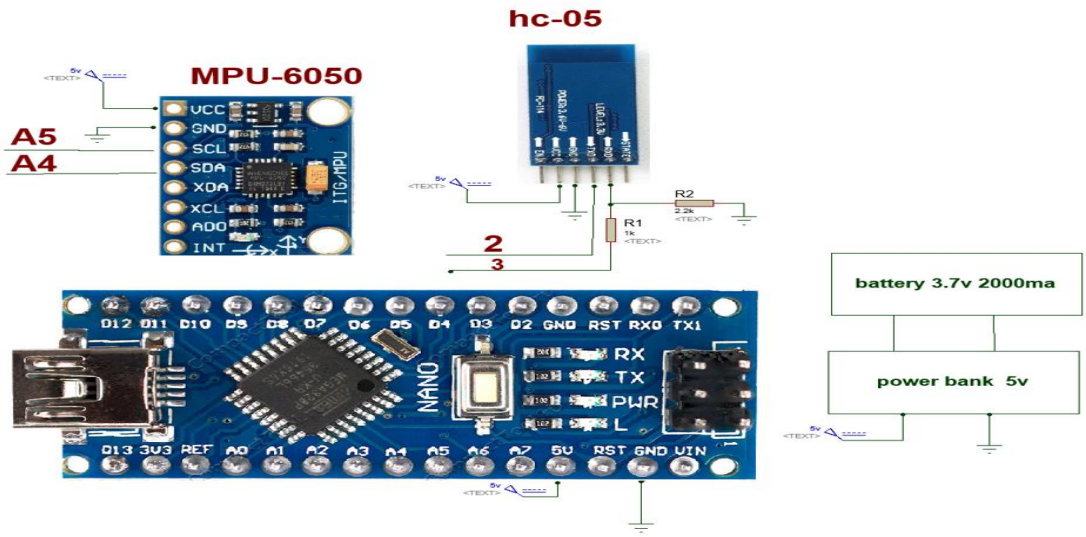


**Figure 3: Transmitter Circuit Diagram**

## 8.Receiver Diagram

The receiver diagram illustrates how the control commands sent from the transmitter unit are received and converted into actual robot motion.

- The **HC-05 Bluetooth module** receives the commands from the transmitter.
- The **Arduino Nano** processes these commands and converts them into control signals for the **L298 motor driver**.
- The L298 controls the **DC motors** to move the robot in the desired directions.
- The system is powered by a **9V battery** to ensure stable operation.
  This diagram helps understand how commands are received, processed, and executed on the robot.

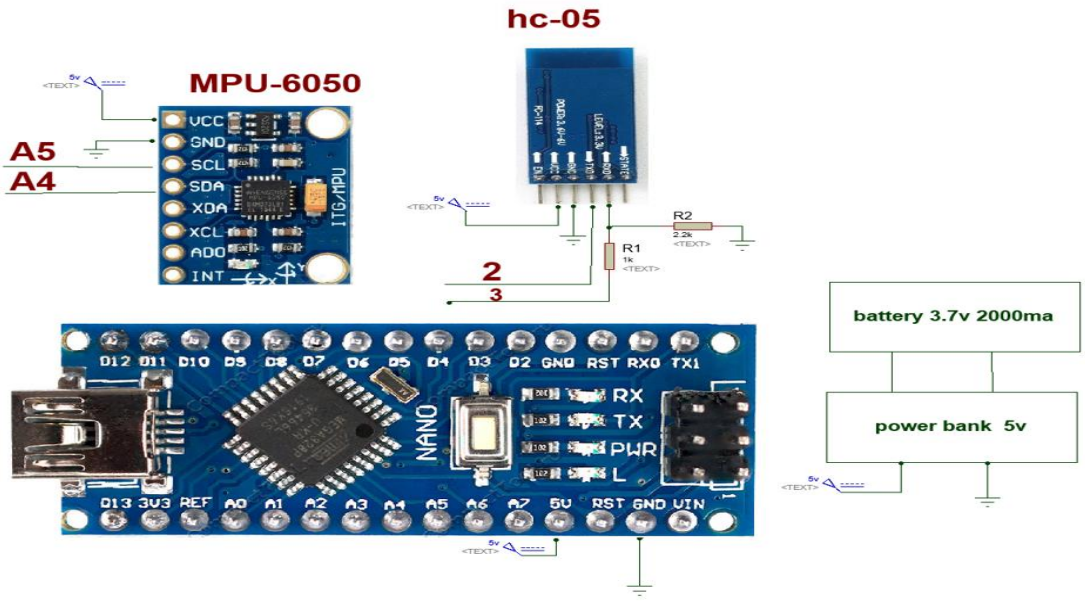



**Figure 4: Receiver Circuit Diagram**

## 9.Technical Specifications

The technical specifications define the essential characteristics that the robot system must meet to ensure the required performance. This includes the robot weight, controller weight, maximum speed, wireless operating range, drive system, motion functions, response time, operating time, and type of wireless communication. These criteria ensure that the robot can perform movements accurately, and that the system operates reliably and stably, providing ease of use and natural interaction through hand gestures.

**Table 1: Technical Specifications of the System**

| Category | Parameter | Specification |
|---|---|---|
| Weight | Robot Weight | ≤ 200 g (Target: 150–200 g) |
| | Controller Weight | ≤ 120 g (Target: 80–120 g) |
| Speed | Maximum Speed | 0.25 – 0.35 m/s (≈ 0.9 – 1.2 km/h) |
| Operating Range | Wireless Range | ≥ 10 m indoor / up to 20–30 m outdoor |
| Drive System | Robot Mobility | Differential drive (2 wheels + caster) |

| | | |
|---|---|---|
| Motion Functions | Movement Modes | Forward – Backward – Left – Right – Stop |
| Sensing System | Motion Detection | Accurate hand motion detection |
| Power System | Power Source | Rechargeable lithium battery |
| Performance | Response Time | < 1 second |
| | Operating Time | 20 – 30 minutes |
| Communication | Wireless Link | Stable wireless connection |

## 10.Components Table

This table lists all the main components used in the robot system, along with the reason for selecting each component. It describes the function of each part and its importance in ensuring efficient operation of the system, including the microcontroller, motion sensors, Bluetooth modules, DC motors, motor driver, and power source. This information helps to understand the system design and justifies the selection of each component.

**Table 2: Components of the System and Reason for Selection**

| Component | Reason for Selection |
|---|---|
| Arduino Nano | was used as the main control unit to process sensor data and send or receive commands for controlling |

| | |
|---|---|
| | the robot's movement. It is characterized by its small size, ease of programming, and low power consumption, making it suitable for portable projects. |
| HC-05 Bluetooth Module | The HC-05 Bluetooth module was used to provide wireless communication between the transmitter and receiver and transmit control commands. It was selected for its easy integration with Arduino and low cost, while Wi-Fi was not used due to higher power consumption. |
| MPU6050 Motion Sensor | The MPU6050 sensor was used to measure hand motion and orientation using an integrated accelerometer and gyroscope. The data are sent to the Arduino for processing and conversion into robot motion commands. It was selected because it combines both functions in a single module with higher accuracy and less complexity than other sensors. |
| L298 Motor Driver Module | The L298 motor driver was used to control the DC motors, acting as an interface between the Arduino and the motors to control their direction and speed. It was selected due to its availability and ease of use compared to other drivers. |
| DC Gear Motors | Gear motors were used because they provide sufficient torque with appropriate speed for robot |

| | |
|---|---|
| | movement, and they are compact and easy to mount. Servo motors were not used as they are designed for angular control, while stepper motors consume more power and require more complex control. |
| Lithium Batteries | Lithium batteries were used as the power source due to their light weight, stable voltage, and high energy density. Alkaline batteries were not used because they provide lower current and discharge quickly when driving motors. |

## 11. Related Work

This table presents the previous studies that were referenced in the development of the project, explaining the principle of each study, the results obtained, and how they were applied in our project.

- The first study (Hand Gesture Controlled Robot Using Arduino) used motion sensors to convert hand gestures into robot commands processed by Arduino, providing a structured approach for gesture recognition and signal processing【1】.
- The second study (Hand Gesture Control Robot) focused on robot control via hand gestures using motion sensors and wireless communication, providing accurate robot direction control and stable response【2】. This table helps to understand the theoretical and practical foundations of the

project and demonstrates how previous study results were leveraged in designing the current system.

**Table 3: Related Work**

| Study | Principle | Results | Application in Our Project |
|---|---|---|---|
| Hand Gesture Controlled Robot Using Arduino | Uses accelerometer/gyroscope to detect hand orientation and convert it into motion commands via Arduino and wireless transmission. | Reliable response, stable wireless communication, and accurate motion control. | Provided a structured approach for gesture recognition and signal processing. |
| Hand Gesture Control Robot | Detects hand movement using motion sensors and converts it into wireless control signals for robot navigation. | Accurate direction control with stable response and communication. | Supported implementation of gesture-based control and motion mapping. |

## 12.Results

The following results were achieved after testing and implementation of the system:

1. **Robot control using hand gestures:** Effective control of the robot was achieved using hand gestures. The robot responded accurately to the commands, including moving forward, backward, turning left and right, and stopping.

    - **Paper No. 1 (Hand Gesture Controlled Robot Using Arduino)** demonstrated the effectiveness of using hand gestures to control the robot, supporting our results from this study.

2. **Fast Response Time:** A quick response time was achieved, ranging from 0.2 to 0.5 seconds between the gesture and the robot's motion, providing a smooth and fast control experience.

    - **Paper No. 2 (Hand Gesture Control Robot)** emphasized the importance of fast response when using hand gestures to control the robot, and our results align with this concept.

3. **Wireless Communication:** A stable Bluetooth connection was achieved between the transmitter and receiver units within a range of 8-10 meters, ensuring reliable remote control.

    - This result aligns with **Paper No. 2** which highlighted the stability of wireless communication and the use of Bluetooth for remote control.

4. **Accuracy of Gesture to Motion Mapping:** Gestures were accurately translated into robot movements, with the robot moving in the specified directions based on the transmitted gestures, with precise directionality such as forward, backward, left, and right.

    - **Paper No. 1** supports the concept of accuracy in executing movements based on gestures, which is also evident in this project.

5. **Robot Movement:** The robot used DC motors with a differential drive system to ensure smooth and stable movement, allowing it to perform the

required tasks accurately.

- This is consistent with techniques used in **Paper No. 1** regarding motor systems and movement control methods.

## 13. Conclusion

This project successfully developed a hand-gesture-based control system for a mobile robot, integrated with wireless control via Bluetooth and all essential electronic components, including the Arduino Nano, MPU-6050 sensor, HC-05 Bluetooth module, L298 motor driver, and DC motors.
The results demonstrated that the system can respond quickly and accurately to gestures, with smooth and stable robot movement and reliable wireless communication within an appropriate range.
This study provides a solid foundation for developing more advanced robots in the future, including integrating a robotic arm, implementing AI-based gesture recognition, and enhancing the user interface for natural interaction. The project also supports the concept of improving human-machine interaction, especially for elderly and physically impaired users.

## 14. References


1. Vala, S. T. (2023). *Hand Gesture Controlled Robot Using Arduino*. Department of Electronics and Communication Engineering, Chandubhai S. Patel Institute of Technology, Changa, India.
2. Gowda, R. V., Sudheer, S. K., Malagamani, S. C., & Dhineshbabu, N. R. (2023). *Hand Gesture Control Robot*. Department of Electronics and Communication Engineering, T. John Institute of Technology, Bengaluru, India.